

DD_SIP-02 - Package Launch Lifecycle

Version: v1.0 **Bundle:** 09_catalogs **Status:** Draft for review **Primary owner:** Catalog / commercial product team **Related DDs:** DD_SIP-01, DD_SIP-04, DD_SIP-05, DD_PLM-CFG-01, DD_PLM-CFG-02, DD_PLM-CFG-03, DD_BIL-CFG-01, DD_EM-CFG-04, FOUNDATION_DROOLS

1. Purpose

SIP-02 defines how SOPHIX moves a package from draft definition to approved, sellable, region/franchise-available, suspended, retired, or end-of-life state.

SIP-01 owns package definition:

- package code;
- service composition;
- base pricing/version records;
- wallet/tax bindings;
- target franchises and technical regions at definition/version level.

SIP-02 owns lifecycle control:

- validation before package can be launched;
- configurable approval before launch;
- activation date and version cutover;
- per-franchise and per-region availability;
- temporary suspension;
- retirement/end-of-life.

Use SIP-02 when:

- product team wants to submit a new package for commercial review;
- a package version must become the current sellable version;
- package availability differs by franchise or region;
- a package must be suspended due to commercial, technical, tax, or provisioning issue;
- a package must stop being sold while existing subscribers remain on it;
- Backoffice, sales app, order capture, or self-care needs the list of sellable packages.

Do not use SIP-02 to change package composition. Composition remains SIP-01 and is locked after activation rules defined in SIP-01.

2. Scope And Boundaries

Capability	Owner
Package master definition and package version records	SIP-01
Package lifecycle transition and launch plan	SIP-02
Service catalog and service active state	PLM-CFG-01
Tax group validity	PLM-CFG-02
Wallet configuration validity	PLM-CFG-03
Billable event compatibility	BIL-CFG-01
Bundle launch and bundle migration	SIP-04
Campaign launch and campaign targeting	SIP-05
Configurable launch approval	EM-CFG-04

Capability	Owner
Configurable launch/availability rules	Drools

Boundary rules:

- SIP-02 changes package/package_version lifecycle state through SIP-01 repository/service contracts or same-module service boundary.
- SIP-02 must not directly modify customer subscriptions. Subscription migration or upgrade is handled by SUB/FUL workflows.
- SIP-02 may make a package unavailable for new sales while existing subscribers remain on their current package version.
- Approval steps are not hardcoded in SIP-02. SIP-02 calls EM-CFG-04. Camunda is used only when the EM-CFG-04 policy mode is BPMN.
- Drools is used for operator-variable launch, readiness, availability, and retirement decisions. Fixed validations remain code/database checks.

3. Lifecycle Overview

```
stateDiagram-v2
    [*] --> DRAFT
    DRAFT --> READY_FOR_REVIEW: validate ok
    READY_FOR_REVIEW --> PENDING_APPROVAL: approval required
    READY_FOR_REVIEW --> APPROVED: auto approved
    PENDING_APPROVAL --> APPROVED: EM-CFG-04 approved
    PENDING_APPROVAL --> REJECTED: EM-CFG-04 rejected
    REJECTED --> DRAFT: revise
    APPROVED --> SCHEDULED: launch date future
    APPROVED --> ACTIVE: launch now
    SCHEDULED --> ACTIVE: activation worker
    ACTIVE --> SUSPENDED: suspend sales
    SUSPENDED --> ACTIVE: resume sales
    ACTIVE --> END_OF_SALE: stop new sales
    END_OF_SALE --> END_OF_LIFE: no active dependency or approved migration plan
    END_OF_LIFE --> RETIRED: archival
```

Master package status in SIP-01 is the high-level state. SIP-02 also manages more granular per-version and per-availability state.

4. State Definitions

State	Meaning	Sellable?
DRAFT	Package exists in SIP-01 but is not validated for launch.	No
READY_FOR_REVIEW	Validation passed and package can enter approval.	No
PENDING_APPROVAL	EM-CFG-04 approval is open.	No
REJECTED	Launch approval rejected; product team must revise.	No
APPROVED	Launch approved but not active yet.	No
SCHEDULED	Launch scheduled for future activation.	No until date
ACTIVE	Package/version is sellable for configured availability.	Yes
SUSPENDED	Temporarily blocked for new sales.	No
END_OF_SALE	Not sellable to new customers; may remain active for existing subscriptions.	No
END_OF_LIFE	Commercially closed; only legacy/historical references remain.	No
RETIRED	Archived and not usable in new flows.	No

5. Data Model

5.1 package_launch_plan

What it is for: stores the launch request for one package version.

When to use it: every time a package version should become sellable, including the first launch and later version price/geo-scope cutover.

Sample meaning: plp-001 launches package pkg-fiber-100m, version pkv-100m-202607, for WIK Nairobi franchises on 2026-07-01.

Column	Type	Purpose	Sample
launch_plan_id	UUID PK	Launch plan id.	plp-001
operator_code	TEXT	Tenant/operator.	WIK
package_id	TEXT	SIP-01 package id.	pkg-01HX_100M
package_code	TEXT	Human-readable package code.	FIBER_100M_RES
package_version_id	TEXT	SIP-01 package_version id.	pkv-100m-202607
launch_type	TEXT	FIRST_LAUNCH, VERSION_CUTOVER, REGION_EXPANSION, RESUME.	FIRST_LAUNCH
requested_launch_at	TIMESTAMP	Desired launch timestamp.	2026-07-01T00:00:00+03:00
effective_timezone	TEXT	Timezone for wall-clock launch.	Africa/Nairobi
status	TEXT	Lifecycle state.	SCHEDULED
approval_policy_code	TEXT NULL	EM-CFG-04 policy code if approval used.	PACKAGE_LAUNCH_APPROVAL
approval_request_id	TEXT NULL	EM-CFG-04 runtime approval request id.	apr-plp-001
approval_mode	TEXT NULL	AUTO, SINGLE_STEP, MULTI_STEP, BPMN.	MULTI_STEP
camunda_process_instance_id	TEXT NULL	Camunda instance id if BPMN-backed.	cam-proc-plp-001
requested_by_user_id	TEXT	Requester user.	usr-product-001
approved_by_user_id	TEXT NULL	Final approver if directly known.	usr-commercial-head
created_at	TIMESTAMP	Creation timestamp.	2026-06-10T09:00:00Z
activated_at	TIMESTAMP NULL	Activation timestamp.	2026-07-01T00:00:05+03:00

Recommended indexes:

- (operator_code, package_id, status).
- (operator_code, requested_launch_at, status).
- (approval_request_id).

5.2 package_launch_check

What it is for: stores launch validation checks and their results.

When to use it: after validation runs, before review/approval, and again just before activation. It explains why launch is blocked or allowed.

Sample meaning: plc-001 says all services inside the package are active in PLM-CFG-01.

Column	Type	Purpose	Sample
check_id	UUID PK	Check id.	plc-001
launch_plan_id	UUID	Launch plan id.	plp-001
check_code	TEXT	Check identifier.	SERVICE_ACTIVE
check_status	TEXT	PASS, WARN, FAIL.	PASS
message	TEXT	Human-readable result.	All package services are active.
source_module_code	TEXT	Module checked.	PLM-CFG-01
source_ref_json	JSONB	Supporting refs.	{"serviceIds":["srv-fiber-100m"]}
checked_at	TIMESTAMP	Check timestamp.	2026-06-10T09:05:00Z

Recommended indexes:

- (launch_plan_id, check_status).
- (check_code, check_status).

5.3 package_availability

What it is for: stores where and through which channels a package version is sellable.

When to use it: sales app, Backoffice order capture, self-care, and API consumers query sellable packages by franchise, region, channel, and date.

Sample meaning: pav-001 makes FIBER_100M_RES available to franchise fr-019 in tech region NRB-WEST for SALES_APP starting 2026-07-01.

Column	Type	Purpose	Sample
availability_id	UUID PK	Availability id.	pav-001
operator_code	TEXT	Tenant/operator.	WIK
package_id	TEXT	Package id.	pkg_01HX_100M
package_version_id	TEXT	Version id.	pkv-100m-202607
franchise_id	TEXT NULL	Franchise scope, null means all allowed.	fr-019
tech_region_code	TEXT NULL	Technical region scope.	NRB-WEST
channel_code	TEXT	BACKOFFICE, SALES_APP, SELF_CARE, API.	SALES_APP
available_from	TIMESTAMP	Start of sellable availability.	2026-07-01T00:00:00+03:00
available_until	TIMESTAMP NULL	End of sellable availability.	null
status	TEXT	SCHEDULED, ACTIVE, SUSPENDED, ENDED.	ACTIVE
reason_code	TEXT NULL	Reason for status.	INITIAL_LAUNCH
created_at	TIMESTAMP	Creation timestamp.	2026-06-10T09:10:00Z
updated_at	TIMESTAMP	Update timestamp.	2026-07-01T00:00:05+03:00

Recommended indexes:

- (operator_code, package_id, status).
- (operator_code, franchise_id, tech_region_code, channel_code, status).
- (operator_code, available_from, available_until).

5.4 package_version_cutover

What it is for: records current-version changes from one package version to another.

When to use it: price changes, geo-scope expansion, tax/wallet binding adjustment, or other SIP-01 version changes that should affect new subscriptions from a date.

Sample meaning: pvc-001 changes new sales from version pkv-100m-202606 to pkv-100m-202607 on July 1. Existing subscriptions stay on their referenced version until a subscription workflow changes them.

Column	Type	Purpose	Sample
cutover_id	UUID PK	Cutover id.	pvc-001
operator_code	TEXT	Tenant/operator.	WIK
package_id	TEXT	Package id.	pkg_01HX_100M
from_version_id	TEXT NULL	Previous active version.	pkv-100m-202606
to_version_id	TEXT	New active version.	pkv-100m-202607
cutover_at	TIMESTAMP	Effective timestamp.	2026-07-01T00:00:00+03:00
status	TEXT	SCHEDULED, COMPLETED, FAILED, CANCELLED.	COMPLETED
launch_plan_id	UUID	Related launch plan.	plp-001

Column	Type	Purpose	Sample
executed_by_worker_id	TEXT NULL	Worker instance.	catalog-worker-01
created_at	TIMESTAMP	Creation timestamp.	2026-06-10T09:12:00Z
completed_at	TIMESTAMP NULL	Completion timestamp.	2026-07-01T00:00:05+03:00

Recommended indexes:

- (operator_code, package_id, cutover_at).
- (status, cutover_at).

5.5 package_lifecycle_event

What it is for: immutable audit trail of package lifecycle changes.

When to use it: every status change, validation result, approval result, activation, suspension, resume, end-of-sale, end-of-life, and retirement.

Sample meaning: ple-001 records that launch plan plp-001 moved package version pkv-100m-202607 to ACTIVE.

Column	Type	Purpose	Sample
event_id	UUID PK	Lifecycle event id.	ple-001
operator_code	TEXT	Tenant/operator.	WIK
package_id	TEXT	Package id.	pkg_01HX_100M
package_version_id	TEXT NULL	Version id.	pkv-100m-202607
launch_plan_id	UUID NULL	Related launch plan.	plp-001
old_status	TEXT NULL	Previous status.	SCHEDULED
new_status	TEXT	New status.	ACTIVE
event_type	TEXT	Lifecycle event type.	PACKAGE_ACTIVATED
reason_code	TEXT	Reason.	SCHEDULED_LAUNCH_REACHED
actor_user_id	TEXT NULL	User actor if manual.	usr-product-001
source_module_code	TEXT	Source module.	SIP-02
metadata_json	JSONB	Extra context.	{"availabilityIds":["pav-001"]}
created_at	TIMESTAMP	Event timestamp.	2026-07-01T00:00:05+03:00

Recommended indexes:

- (operator_code, package_id, created_at DESC).
- (launch_plan_id, created_at DESC).

5.6 package_retirement_plan

What it is for: stores controlled end-of-sale/end-of-life plans.

When to use it: product team wants to stop selling a package, retire a package version, or remove a package after existing subscribers have migrated.

Sample meaning: prp-001 stops new sales on 2026-09-01 but keeps existing customers active until migration completes.

Column	Type	Purpose	Sample
retirement_plan_id	UUID PK	Retirement plan id.	prp-001
operator_code	TEXT	Tenant/operator.	WIK
package_id	TEXT	Package id.	pkg_01HX_0LD50
package_version_id	TEXT NULL	Optional version id.	pkv-0ld50-202601
retirement_type	TEXT	END_OF_SALE, END_OF_LIFE, RETIRE_VERSION.	END_OF_SALE
effective_at	TIMESTAMP	Effective timestamp.	2026-09-01T00:00:00+03:00
existing_subscriber_policy	TEXT	KEEP_AS_IS, MIGRATE_REQUIRED, BLOCK_RENEWAL.	KEEP_AS_IS

Column	Type	Purpose	Sample
migration_workflow_ref	TEXT NULL	SUB/FUL workflow if migration needed.	SUB-WF-MIGRATION-01
approval_request_id	TEXT NULL	EM-CFG-04 approval id if required.	apr-prp-001
status	TEXT	DRAFT, PENDING_APPROVAL, APPROVED, SCHEDULED, COMPLETED, CANCELLED.	SCHEDULED
created_by_user_id	TEXT	Requester.	usr-product-001
created_at	TIMESTAMP	Creation timestamp.	2026-08-15T09:00:00Z
completed_at	TIMESTAMP NULL	Completion timestamp.	2026-09-01T00:00:04+03:00

Recommended indexes:

- (operator_code, package_id, status).
- (status, effective_at).

6. APIs

6.1 Create Launch Plan

POST /api/package-launch-plans
 Idempotency-Key: package-launch-FIBER_100M_RES-pkv-202607
 Content-Type: application/json

```
{
  "operatorCode": "WIK",
  "packageId": "pkg_01HX_100M",
  "packageCode": "FIBER_100M_RES",
  "packageVersionId": "pkv-100m-202607",
  "launchType": "FIRST_LAUNCH",
  "requestedLaunchAt": "2026-07-01T00:00:00+03:00",
  "effectiveTimezone": "Africa/Nairobi",
  "availability": [
    {
      "franchiseId": "fr-019",
      "techRegionCode": "NRB-WEST",
      "channelCode": "SALES_APP"
    },
    {
      "franchiseId": "fr-019",
      "techRegionCode": "NRB-WEST",
      "channelCode": "BACKOFFICE"
    }
  ]
}
```

Response:

```
{
  "launchPlanId": "plp-001",
  "status": "DRAFT"
}
```

6.2 Validate Launch Plan

POST /api/package-launch-plans/{launch_plan_id}/validate

SIP-02 checks:

- SIP-01 package exists and version exists;
- package is in a launchable state;
- package composition services are active in PLM-CFG-01;
- tax group is valid in PLM-CFG-02;
- wallet configuration is valid in PLM-CFG-03;
- billing event compatibility exists in BIL-CFG-01;
- requested availability matches package target franchise/region restrictions;

- Drools launch readiness rules pass or warn.

6.3 Submit Launch Plan For Approval

```
POST /api/package-launch-plans/{launch_plan_id}/submit-review
Content-Type: application/json
```

```
{
  "requesterUserId": "usr-product-001",
  "comment": "Ready for Nairobi launch."
}
```

SIP-02 calls EM-CFG-04:

```
POST /api/approvals/evaluate
Content-Type: application/json
```

```
{
  "operatorCode": "WIK",
  "moduleCode": "SIP-02",
  "actionCode": "PACKAGE_LAUNCH_APPROVAL",
  "subjectType": "PACKAGE_LAUNCH_PLAN",
  "subjectId": "plp-001",
  "context": {
    "launchType": "FIRST_LAUNCH",
    "packageCode": "FIBER_100M_RES",
    "requestedLaunchAt": "2026-07-01T00:00:00+03:00",
    "channelCodes": ["SALES_APP", "BACKOFFICE"],
    "franchiseCount": 1,
    "techRegionCount": 1
  }
}
```

If approval is required, SIP-02 starts an approval request through EM-CFG-04 and stores approval_request_id. Approvers act through EM-CFG-04 task APIs.

6.4 Approval Outcome Callback

```
POST /internal/package-launch-plans/{launch_plan_id}/approval-outcome
Content-Type: application/json
```

```
{
  "approvalRequestId": "apr-plp-001",
  "outcome": "APPROVED",
  "decisionComment": "Approved for WIK Nairobi launch."
}
```

If approved, SIP-02 moves the plan to APPROVED or SCHEDULED. If rejected, status becomes REJECTED.

6.5 Activate Launch Plan

```
POST /api/package-launch-plans/{launch_plan_id}/activate
```

Manual activation is allowed only if:

- launch is approved;
- requested launch time is now or past;
- final validation has no FAIL;
- package version has not been superseded or cancelled.

Scheduled activation should be performed by a worker.

6.6 Get Available Packages

Used by Backoffice, sales app, order capture, self-care, and API consumers.

```
GET /api/packages/available?operatorCode=WIK&channelCode=SALES_APP&franchiseId=fr-019&techRegionCode=NRB-WEST&onDate=2026-
```

Response:

```
{
  "packages": [
    {
      "packageId": "pkg_01HX_100M",
      "packageCode": "FIBER_100M_RES",
      "packageVersionId": "pkv-100m-202607",
      "displayName": "Fiber 100M Residential",
      "price": 3000.00,
    }
  ]
}
```

```

        "currency": "KES",
        "billingFrequencyDays": 30,
        "availableChannels": ["SALES_APP", "BACKOFFICE"]
    }
}
}

```

6.7 Suspend Package Availability

POST /api/packages/{package_id}/availability/suspend
Content-Type: application/json

```

{
  "packageVersionId": "pkv-100m-202607",
  "franchiseId": "fr-019",
  "techRegionCode": "NRB-WEST",
  "channelCode": "SALES_APP",
  "reasonCode": "TEMPORARY_PROVISIONING_ISSUE",
  "comment": "ONT stock shortage in Nairobi West."
}

```

6.8 Resume Package Availability

POST /api/packages/{package_id}/availability/resume
Content-Type: application/json

6.9 Create Retirement Plan

POST /api/package-retirement-plans
Idempotency-Key: package-retire-OLD50-20260901
Content-Type: application/json

```

{
  "operatorCode": "WIK",
  "packageId": "pkg_01HX_OLD50",
  "retirementType": "END_OF_SALE",
  "effectiveAt": "2026-09-01T00:00:00+03:00",
  "existingSubscriberPolicy": "KEEP_AS_IS",
  "reasonCode": "PRODUCT_REPLACED_BY_100M"
}

```

7. Business Rules

Rule	Description	Reason
R-SIP-02-01	A package version can be launched only if package and version exist in SIP-01.	Prevent orphan launch.
R-SIP-02-02	Package composition services must be active in PLM-CFG-01.	Prevent unsellable package.
R-SIP-02-03	Wallet and tax configuration must be valid for operator/currency/region.	Billing correctness.
R-SIP-02-04	Availability scope must be inside package version target franchises/regions.	Commercial control.
R-SIP-02-05	Launch approval must be evaluated through EM-CFG-04.	Shared approval governance.
R-SIP-02-06	Camunda is used only when EM-CFG-04 approval mode is BPMN.	Avoid unnecessary workflow complexity.
R-SIP-02-07	Activation must run final validation immediately before status changes.	Avoid stale launch checks.
R-SIP-02-08	Activating a new version updates package current_version_id for new sales only.	Protect existing subscriptions.
R-SIP-02-09	END_OF_SALE blocks new sales but does not terminate existing subscriptions.	Customer continuity.
R-SIP-02-10	Retirement requiring customer migration must reference SUB/FUL migration workflow.	Avoid orphan customers.
R-SIP-02-11	Per-franchise/channel availability can be suspended without changing master package status.	Localized operational control.
R-SIP-02-12	Package activation must publish events for frontend/API cache refresh.	Prevent stale sales options.

8. Rule Engine Guidance

Use Drools for configurable launch and availability decisions:

Rule file	Purpose
package-launch-readiness.drl	Adds operator-specific validation, warnings, and hard stops before approval.
package-launch-approval.drl	Decides approval policy code or auto-approval context before EM-CFG-04 call.
package-availability.drl	Controls channel/franchise/region availability where policy is not a simple DB filter.
package-retirement-policy.drl	Decides whether end-of-sale/end-of-life requires approval or migration plan.

Example:

```
package com.sophix.catalog.package_lifecycle

rule "First launch to sales app requires commercial approval"
when
    $p : PackageLaunchFact(launchType == "FIRST_LAUNCH", channelCodes contains "SALES_APP")
    $d : PackageLaunchDecision()
then
    $d.setApprovalRequired(true);
    $d.setApprovalPolicyCode("PACKAGE_LAUNCH_COMMERCIAL");
    $d.addReason("FIRST_LAUNCH_SALES_APP");
end
```

Fixed checks such as package existence, version existence, required fields, and idempotency remain normal application validation.

9. Events

Event	Topic	When emitted	Consumers
PackageLaunchPlanCreated	sophix.catalog.package.launch-plan-created	Launch plan created.	Backoffice, REP.
PackageLaunchValidated	sophix.catalog.package.launch-validating	Validation completed.	Backoffice.
PackageLaunchApprovalRequired	sophix.catalog.package.launch-approval-required	EM-CFG-04 request started.	Backoffice, ICN.
PackageLaunchApproved	sophix.catalog.package.launch-approved	Approval completed.	Backoffice, REP.
PackageActivated	sophix.catalog.package.activated	Package/version becomes active.	FUL-02, sales app cache, Backoffice, REP.
PackageAvailabilityChanged	sophix.catalog.package.availability-changed	Availability added/suspended/resumed/ended.	FE apps, FUL-02, API cache.
PackageEndOfSale	sophix.catalog.package.end-of-sale	Package no longer sellable.	Sales apps, order capture.
PackageRetired	sophix.catalog.package.retired	Package/version archived.	REP, audit.

10. Frontend And API Consumers

Frontend area	APIs/modules used	Notes
Product Backoffice	SIP-01 package APIs plus SIP-02 launch APIs	Create package in SIP-01, launch it through SIP-02.
Approval inbox	EM-CFG-04 task APIs	Launch approval tasks are not local SIP-02 tasks.
Backoffice order capture	GET /api/packages/available	Lists sellable packages for selected customer location/channel.
Sales outdoor app	GET /api/packages/available through FUL-02 quote flow	Must filter by franchise, territory, HomePass/region, and channel.
Self-care	GET /api/packages/available with customer/subscription context	Only eligible package changes shown.
Reporting	SIP-02 events and read models	Launch pipeline, availability, package lifecycle KPIs.

11. Worker Model

Workers can be consolidated with catalog/commercial workers.

Worker job	Purpose	Frequency
package-launch-activation	Activates approved scheduled launch plans when due.	Every minute or event-driven.
package-cutover-runner	Updates current version and supersedes old version.	Every minute or due queue.
package-retirement-runner	Executes approved end-of-sale/end-of-life plans.	Every 5 minutes.
package-lifecycle-outbox-publish	Publishes lifecycle events.	Continuous/short interval.
package-availability-cache-refresh	Refreshes read cache after lifecycle changes.	Event-driven.

These are command/job types, not necessarily separate deployments.

12. Infrastructure Recommendations

For a medium budget deployment, SIP-02 should run inside the catalog/commercial API deployment together with SIP-01/SIP-03/SIP-04/SIP-05 unless load proves otherwise.

Component	Recommendation
API	Same catalog API service exposes /api/package-launch-plans, /api/packages/available, and retirement APIs.
Database	Catalog schema in PostgreSQL.
Cache	Redis for package availability read model, invalidated by lifecycle events.
Workers	Shared catalog worker deployment.
Rules	Use existing Drools/KIE foundation.
Approval	Use EM-CFG-04 and existing Camunda only for BPMN-backed policies.

Suggested medium starting size:

Layer	Starting size
Catalog API pods	2 replicas, 2 vCPU, 4 GB RAM each.
Catalog workers	1 replica, 2 vCPU, 4 GB RAM.
PostgreSQL	Shared catalog/business DB; no separate DB for SIP-02.
Redis	Shared cache namespace catalog:package-availability.

13. AWS Deployment Notes

AWS layer	Recommendation
Compute	ECS/EKS service colocated with catalog API and worker.
Database	RDS PostgreSQL, same catalog/business DB cluster.
Cache	ElastiCache Redis for availability cache.
Events	MSK or Kafka-compatible broker using catalog topics.
Secrets	AWS Secrets Manager for DB, service tokens, KIE, and EM-CFG-04 credentials.
Scheduling	EventBridge can trigger worker tasks if not using internal scheduler.

Keep package availability cache small and invalidation-driven. A stale sales package list directly affects customer onboarding.

14. Testing Guidance

Test	Expected result
Launch package without existing SIP-01 version	Validation fails.

Test	Expected result
Launch package with inactive service	Validation FAIL, cannot submit review.
Launch requiring approval	EM-CFG-04 request created; SIP-02 waits for callback.
EM-CFG-04 rejection	Launch plan becomes REJECTED, package remains not sellable.
Scheduled activation due	Package/version becomes active and availability rows become active.
Activation final validation fails	Launch remains blocked and event/alert created.
Available packages query by franchise/channel	Only active matching package versions returned.
Suspend availability	Sales app no longer sees package for that scope.
End of sale	New sales blocked, existing subscription references remain valid.
Version cutover	current_version_id changes for new sales only.

15. Acceptance Criteria

- SIP-02 launch plan, check, availability, cutover, lifecycle event, and retirement tables exist.
- Package launch validates SIP-01, PLM-CFG, and BIL-CFG dependencies.
- Launch approval is routed through EM-CFG-04; SIP-02 does not hardcode approval steps.
- Drools rule files exist for configurable readiness, approval, availability, and retirement policy.
- Activation updates package/version current state for new sales without changing existing subscriptions.
- GET /api/packages/available is usable by Backoffice, sales app, order capture, and self-care.
- Lifecycle and availability events are emitted through outbox.
- Tests cover validation, approval callbacks, activation, suspension, end-of-sale, and version cutover.